

Carter Reid

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EDUCATION

University of Waterloo | Bachelor of Computer Science (3A CO-OP)

Sept. 2023 - Present

EXPERIENCE

Backend Developer | CIBC

September 2025 – May 2026

- Designed and developed reusable backend libraries and API infrastructure in Go, Docker, and OpenShift to accelerate development of secure internal payment services
- Built a networking framework for internal payment APIs, including concurrency, asynchronous processing, retries, failover handling, throttling, and distributed service communication
- Developed a SAF (Store and Forward) system for CIBC payment APIs serving millions of clients and supporting traffic up to 600 TPS, implementing persistent queueing and replay mechanisms using Azure-hosted PostgreSQL alongside OAuth2 authentication and JWT token validation
- Created stress testing infrastructure, mock clients, and observability tooling while optimizing latency, throughput, CPU usage, and reliability under high traffic spikes
- Managed containerized service deployment and artifact distribution workflows using Docker, OpenShift, and JFrog Artifactory

Software Developer | Tuq

Jan 2025 – May 2025

- Developed features for a pharmacy web platform, including dynamic forms, configurable question flows, and subscription management functionality
- Integrated an AI chatbot into an iOS and Android mental wellness app with contextual chat history, voice recognition, and text-to-speech support
- Built reusable React components and collaborated across 6+ concurrent software projects in an agile development environment

Sales Marketing Coordinator | Electrolux

Jun 2024 – Aug 2024

- Developed a program to automatically populate emails with various product details from a database using Python and Pandas library
- Created a VBA Script to automate the sending of thousands of unique emails to multiple different addresses
- Generated business interest and brand awareness through a large-scale email marketing campaign targeting over a thousand developers, thousands of construction projects, and over half a million living units

PROJECTS

Theatre VTT | TypeScript, React, WebSockets, Node.js

2025

- Designed a server-authoritative real-time synchronization system using WebSockets to maintain consistent multiplayer game state across connected clients
- Designed an event-driven rule processing engine capable of resolving chained combat interactions, reactions, conditions, and automated state transitions
- Implemented persistent storage and session recovery systems for multiplayer game states, user-generated content, and customizable scripting functionality for game actions and dice-based commands

Ritual | C++, Parsing, Systems Programming

2025

- Designed and implemented a custom pseudocode-based scripting language for OS-level task automation and event-driven workflows
- Built a trigger execution system capable of running scripts on scheduled intervals and system events such as application closure, startup, and file downloads
- Implemented conditional execution and filesystem operations enabling automated file organization and rule-based task pipelines

TECHNICAL SKILLS

Languages: Go, C, C++, C#, Java, Python, SQL, Bash

Backend & Infrastructure: REST APIs, WebSockets, Distributed Systems, Async Processing, Goroutines, OAuth2, JWT, Docker, OpenShift, Node.js

Databases & Observability: PostgreSQL, AzureDB, Splunk, Persistent Storage, Metrics, Logging, Load Testing

Developer Tools: Git, GitHub, VS Code, WebStorm, RubyMine, Visual Studio, Confluence